

Homework (Carolina Reaper)

- Q1.** If $f(x) = x \cdot 2$ and $g(x) = x + 3$, what is $(f + g)(x)$?
(A) $x \cdot 2 + x + 3$
(B) $x \cdot 2 - x - 3$
(C) $x + 3 \cdot 2$
(D) $x - 3 \cdot 2$
(E) $x \cdot 2 - x + 3$
- Q2.** What is the domain of $(f/g)(x)$ if $f(x) = x^2$ and $g(x) = x - 4$?
(A) $(-\infty, 4) \cup (4, \infty)$
(B) $(-\infty, 4)$
(C) $[4, \infty)$
(D) $(-\infty, 4] \cup [4, \infty)$
(E) $[4, \infty)$
- Q3.** The rhythm pattern of a song is defined by $f(x) = \sin(x)$ and $g(x) = \cos(x)$. What is $(f \cdot g)(x)$?
(A) $\sin(x) \cdot \cos(x)$
(B) $\sin(x) + \cos(x)$
(C) $\sin(x) - \cos(x)$
(D) $\sin(x)/\cos(x)$
(E) $\cos(x)/\sin(x)$
- Q4.** If $h(x) = (f/g)(x)$ and $f(x) = x + 2$ and $g(x) = x - 1$, what is $h(3)$?
(A) $5/2$
(B) $2/5$
(C) $1/2$
(D) $3/2$
- (E) $7/2$
- Q5.** What is the simplified expression of $(x^2 + 3) \cdot (x - 2)$?
(A) $x^3 - 2x^2 + 3x - 6$
(B) $x^3 - 2x^2 - 3x + 6$
(C) $x^3 + 2x^2 + 3x + 6$
(D) $x^3 - 2x^2 - 3x - 6$
(E) $x^3 + 2x^2 - 3x - 6$
- Q6.** If $f(x) = x^2$ and $g(x) = x - 1$, what is $(f - g)(x)$?
(A) $x^2 - x + 1$
(B) $x^2 + x - 1$
(C) $x^2 - x - 1$
(D) $x^2 + x + 1$
(E) $x^2 - 1$
- Q7.** What is the domain of $(f + g)(x)$ if $f(x) = 1/x$ and $g(x) = 2/x$?
(A) $(-\infty, 0) \cup (0, \infty)$
(B) $(-\infty, 0)$
(C) $[0, \infty)$
(D) $(-\infty, 0] \cup [0, \infty)$
(E) $(-\infty, 0] \cup (0, \infty)$
- Q8.** If $f(x) = x + 1$ and $g(x) = x - 2$, what is $(f/g)(x)$?
(A) $(x + 1)/(x - 2)$
(B) $(x - 2)/(x + 1)$
(C) $x + 1 + x - 2$
(D) $x + 1 - x + 2$
(E) $x - 2 - x - 1$

Open Ended Questions (FRQ)

- Q9.** The canvas size of a painting is defined by $f(x) = x + 2$ and $g(x) = 2x - 1$. Determine the domain of $(f \cdot g)(x)$ and simplify the resulting expression. Show all steps and calculate the value of $(f \cdot g)(2)$.
- Q10.** The frequency of a sound wave is defined by $h(x) = (f/g)(x)$ where $f(x) = \sin(x)$ and $g(x) = \cos(x)$. Determine the domain of $h(x)$ and simplify the resulting expression. Show all steps and calculate the value of $h(\pi/4)$.

Chef's Tips

- Tip 1: Ensure students provide the correct units for their answers.
- Tip 2: Require students to show all steps for free response questions.
- Tip 3: Remind students to check their work for accuracy and completeness.